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PRACTICE INSIGHTS

Locations and management strategies for prioritizing King Rail conservation efforts in the Mississippi Alluvial Valley

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Abstract

1. King Rail populations, like those of many wetland-dependent waterbirds, have declined throughout much of their range due primarily to habitat loss and degradation. The Mississippi Alluvial Valley (MAV) is a broad region where King Rails have declined precipitously.
2. In this practical guide to King Rail conservation, we used results from an existing Bayesian network (BN) analysis to predict where King Rails are most likely to occur in the MAV. Then, we identified 28 properties with a conservation focus in this region where we predicted that King Rail habitat management efforts would be most effective.
3. Although managers of most of these properties may not focus on King Rail habitat, management efforts at a wetland complex in the MAV have already resulted in what appears to be high-quality King Rail habitat. We describe the management activities that have been implemented at this property to provide guidance for others interested in creating conditions amenable to King Rail use.
4. We also present opinions of King Rail experts who contributed to our BN models and describe how current knowledge could be used to achieve conservation goals meant to benefit King Rails and other waterbirds throughout the MAV.
5. *Practical implication:* Our results could assist efforts to identify, improve and/or create habitat for King Rails.

KEYWORDS
Bayesian network, conservation, expert opinion, habitat, King Rail, *Rallus elegans*

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More than 50% of wetlands have been lost in the United States since European colonization (Tiner, 1984) and this loss is likely a cause of the >20% decline in waterbird abundance in North America since 1970 (Rosenberg et al., 2019). One waterbird species that has experienced some of the most substantial population declines is the King Rail (*Rallus elegans*; Hostetler et al. 2023). Although King Rails occur across most of the eastern United States, their populations have declined by approximately 90% since 1966 (Hostetler et al. 2023). The King Rail has a state listing indicating this population decline, such as 'threatened' or 'endangered', in nearly every state where the species occurs (Cooper, 2008), including many of those within the Mississippi Alluvial Valley (MAV). The MAV region includes the alluvial plain of the Mississippi River in the following states: Illinois, Missouri, Kentucky, Tennessee, Arkansas, Mississippi and Louisiana. Similar to many other regions within the range of the King Rail, there are few remaining locations in the MAV where this species is known to breed. For example, in Arkansas, the only documented location where King Rails are known to reproduce is the 'west unit' of the Freddie Black Choctaw Island Wildlife Management Area/Deer Research Area (Hargrove & Osborne, 2023; Choctaw West). Developing and implementing effective conservation strategies in regions such as the MAV, where most King Rail habitat has been lost or degraded, could help to prevent further declines and potentially increase King Rail populations (Roberts et al., 2025).

King Rails in the MAV are considered by the Lower Mississippi Valley Joint Venture to be a 'priority species' that would likely benefit from efforts to create or improve habitat (LMVJV, 2023). These management efforts would probably have a positive impact on other wetland species that have also declined due to dependence upon shallow water (<20 cm) marsh conditions (Budd & Krementz, 2011; Malone et al., 2023; Meanley, 1969). An increase in King Rail populations, and those of other wetland-dependent species that have similar habitat requirements, could potentially be facilitated by identification of areas where management efforts are likely to be most effective and by a description of management practices that can create high-quality habitat at these locations. Therefore, in this practical guide, we used existing models of King Rail occupancy in the MAV (Brewer et al., 2025) to identify locations where management actions could be most effective. We also describe management practices at a property in the MAV where conservation practitioners have successfully created King Rail habitat. Finally, we present common themes of King Rail ecology and habitat use based on discussions with individuals who had expertise (e.g. wetland managers) regarding King Rail ecology and management.

WHERE TO PRIORITIZE KING RAIL CONSERVATION

Bayesian networks (BNs) are statistical tools that can incorporate information from a variety of sources, including empirical data and expert knowledge. BNs can be particularly useful for wildlife managers because of their ability to predict outcomes of management actions by using relationships among variables represented visually in a model that can be easily updated when new information becomes available (Smith et al., 2007). A recent modelling effort used BNs to predict King Rail occupancy throughout the MAV during both breeding and non-breeding periods (Brewer et al., 2025). The goal of that study was to compare the effectiveness of two different modelling approaches. One modelling approach involved experts taking part in a double-blind peer-review process to produce BNs (PRBN) and the other relied primarily on a literature review to produce BNs (LRBN). Although uncertainty regarding the probability of King Rail occupancy could likely be reduced with additional field data, both the LRBN and PRBN models were effective at predicting King Rail occupancy during the breeding (15 March to 15 August) and non-breeding (16 August–14 March) periods (Brewer et al., 2025). These models produced similar, but different, estimates of King Rail occupancy throughout the MAV (Figure S1). Considered together, the LRBN and PRBN models could assist in prioritizing locations for King Rail conservation efforts and are available to suit the various sensitivity and specificity needs of practitioners (refer to supplementary .asc files). Herein, we identify conservation-focused properties where these models predicted the highest probability of King Rail occupancy (a conservative approach, minimizing false positives) based on home range and landscape-scale characteristics. Brewer et al. (2025) provides full model documentation for the LRBN and PRBN models.

We began by evaluating alignment between localities where each BN model predicted a high probability of King Rail occupancy and where individuals of this species occurred according to satellite telemetry data, which were derived from Schmit (2024). The LRBN model performed better than the PRBN model (refer to Supporting Information), for both the breeding and non-breeding periods, and so was our primary tool for identifying King Rail habitat. We also used the PRBN to fine-tune conclusions, given this model's ability to minimize the false positive rate for occupancy predictions in some circumstances (Brewer et al., 2025).

We identified pixels (30 m × 30 m), for both the breeding and non-breeding period, where predicted probability of King Rail occupancy matched or exceeded 99% of the pixels in the MAV according to the LRBN model. We assumed that these areas possessed potential high-quality habitat (hereafter: 'habitat'), where efforts to create and/or conserve conditions upon which King Rails rely would be most effective. We used the Protected Areas Database (USGS GAP, 2024) to identify properties, such as National Wildlife Refuges, that are owned (often publicly) and managed with the intention of conserving natural communities. Then, we separately quantified the amount of habitat at each of these properties according to the LRBN for the breeding and non-breeding periods. We also quantified the amount of habitat within each property, and during each period, that both the LRBN and the PRBN predicted.

Consequently, we identified 28 properties (Table 1) that contained ≥80 ha of King Rail breeding habitat, an area that could support ≥5 non-overlapping King Rail home-ranges (~16 ha; Brewer

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TABLE 1 Overview of properties in the Mississippi Alluvial Valley where the primary Bayesian network (BN) model predicted ≥80 ha of King Rail breeding habitat. At each property, we also quantified the amount of predicted non-breeding habitat. A 'Br' in the 'Identified by Both Models?' column indicates that both the primary and secondary BN identified the same potential habitat patches which spanned ≥80 ha at the associated property during the breeding period; a 'Nb' indicates the same for the non-breeding period; a '–' indicates the primary and secondary BNs did not identify ≥80 ha of the same habitat at the associated property during either period.

State	Owner	Property[a]	Breeding habitat (ha)	Non-breeding habitat (ha)	Ident. By both models?
AR	Federal	Dale Bumpers White River NWR	3183	14,275	–
MO	Federal	Mingo NWR	1299	1542	Br
LA	State	LSU Research Station	954	967	Nb
MS	Federal	Coldwater River NWR	812	928	–
AR	State	Saint Francis Sunken Lands WMA	798	None	–
AR	Federal	Cache River NWR	784	6393	–
AR	State/Fed	Trusten Holder WMA	587	809	–
MO	State	Otter Slough CA	585	886	–
MO	State	Duck Creek CA	464	911	Br
TN/KY	Federal	Reelfoot NWR	405	836	–
MS	Federal	St. Catherine Creek NWR	383	1714	Br, Nb
MO	State	Little River CA	367	None	–
LA	State	Red River WMA	354	406	Nb
AR	Federal	Big Lake NWR	333	None	–
LA	Federal	Grand Cote NWR	268	1077	Nb
LA	State	Lake Boeuf WMA	249	248	Nb
KY	State/Fed	Boatwright WMA	218	349	–
AR	State	Steve N. Wilson/Raft Creek Bottoms WMA	177	1431	Br
LA	NGO	Cypress Island	126	985	–
MO	State	Coon Island CA	117	489	–
AR	Federal	Arkansas Post National Memorial	117	167	–
MO	State	Duck Creek CA (Dark Cypress Swamp Unit)	117	None	–
MO	State	Hornersville Swamp CA	116	179	–
LA	State	Three Rivers WMA	90	None	Br
TN	State	Black Bayou Refuge	89	132	–
AR	Unknown	Willow Beach Rec. Area	88	None	–
MO	State	Ten Mile Pond CA	87	216	–
KY	State	Obion Creek WMA	86	476	–

Abbreviations: CA, conservation area; NGO, non-governmental organization; NWR, national wildlife refuge; WMA, wildlife management area.
[a]Unnamed properties owned by Louisiana (e.g. Division of State Lands) and Mississippi (16th Section Public School Trust Lands) also contained ≥80 ha of King Rail habitat.

et al., 2023). These properties were dispersed across the MAV in Arkansas ($n=8$), Missouri ($n=8$), Louisiana ($n=6$), Kentucky ($n=3$), Tennessee ($n=2$) and Mississippi ($n=2$) and included land owned by federal, state, local and non-governmental entities (Table 1). St. Catherine Creek National Wildlife Refuge, in Mississippi, was one of five properties (Table 1) where the LRBN and PRBN predicted ≥80 ha of breeding habitat at the same location (i.e. same pixels) and was the only property where both models predicted ≥80 hectares of non-breeding habitat at the same location. Based on our models, environmental conditions at this property could serve as a model for King Rail conservation efforts elsewhere. We identified several properties that contained ≥80 ha of habitat, but which occurred primarily in the coastal region of Louisiana and only partially in the MAV. These properties, which are not included in Table 1, are Maurepas Swamp Wildlife Management Area, Jean Lafitte National Historical Park and Preserve, Mandalay Bay National Wildlife Refuge and Bayou Teche National Wildlife Refuge. Although other properties likely possess conditions suitable for King Rails, those that we identified could be prioritized during efforts to maintain and/or create additional habitat for this species.

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A CASE STUDY IN MANAGING WETLANDS TO BENEFIT KING RAILS

Choctaw West is a property in southeastern Arkansas where ongoing management efforts to create habitat for King Rails have been successful. Thus, this property's history and an overview of the management practices implemented there could inform King Rail habitat management elsewhere in the MAV. In the mid-1990s, this ~730-ha, privately owned area was removed from agricultural production upon enrolment in the Wetland Reserve Easement (WRE) programme. In 2006, King Rails were detected at this property during standardized marsh bird surveys (Budd & Krementz, 2011). The Arkansas Game and Fish Commission (AGFC) purchased Choctaw West in 2008. No King Rails were detected during secretive marsh bird surveys conducted in 2012 (J. Novobilsky, AGFC, written communication, 2025), potentially due to encroachment by woody vegetation. In 2014, the AGFC and the Natural Resources Conservation Service (NRCS) began restoration efforts to remove trees and create impoundments to support greater diversity within robust emergent plant communities and varied water depths needed to promote secretive marsh bird habitat.

AGFC employees began to conduct annual standardized marsh bird surveys at Choctaw West in 2015. The number of King Rails detected increased over time, ranging from 6 in 2015 to a high of 38 in 2020 (J. Novobilsky, AGFC, written communication, 2025). Additionally, researchers banded 20 individuals in 2022 and 35 individuals in 2023 during the effort that generated the King Rail satellite telemetry locations (Schmit, 2024) used to evaluate the BN models associated with our study. Nesting success has been confirmed at Choctaw West (Hargrove & Osborne, 2023; Schmit, 2024), which suggests ongoing management efforts are achieving a primary management goal at the site: to create and maintain high-quality King Rail habitat. This goal appears compatible with supporting waterfowl populations, which are also considered when management decisions are made at Choctaw West, given that many diving and dabbling ducks are observed during both migratory and non-breeding periods.

Water management is a component of year-round efforts to create appropriate conditions for King Rails (Meanley, 1969) and other waterbirds at Choctaw West. Water is supplied to the seven marsh units at the property via two surface water relift pumps. The topography of these marsh units has elevation changes that provide a range of water depths. Water levels are typically deepest during the winter months to provide habitat for waterfowl. Our best estimate of water depths where King Rails occur during the winter months is 30–50 cm (J. McCallie, personal observation). In the spring, a slow drawdown is achieved by releasing water through numerous water control structures in the marsh units. By early summer, the water depths average ~20 to ~36 cm in areas that King Rails primarily use. Water is pumped into marsh units each summer during drier periods and allowed to remain for several days to ensure that these units do not completely dry. Then, most of the water is slowly released into adjacent units through water control structures. This flush and drain technique provides many small pockets of water that provide optimum feeding habitat for juvenile King Rails (Darrah & Krementz, 2011). From mid-summer to early fall, water depths of 10–30 cm are targeted (J. McCallie, AGFC, written communication, 2025).

The targeted plant community at Choctaw West is dominated by herbaceous wetland plants and is achieved through a variety of mechanical and chemical treatments. In late summer and early autumn, a tractor and a cleated rice stubble roller are used to roll, chop and turn soil in paths and patches throughout the marsh units (Figure S2). It is advantageous to have some standing water when this treatment is implemented to achieve conditions resembling a mudflat. This rolling technique is meant to increase plant species and structural diversity, which requires a unit-based strategy informed by topography, water depth/inundation period and the existing plant community. The strategy involves rolling over plant species that may be less desirable (e.g. hemp sesbania [*Sesbania herbacea*] and water smartweed [*Persicaria ampibhia*]) or more common in a certain unit and encouraging the growth of more desirable (e.g. square stem spike rush [*Eleocharis quadrangulata*]) or less common species in the same unit. A greater number of passes made with the roller increases soil disturbance, thereby maximizing the persistence of the created open water area before vegetation returns. Chemical applications, disking and prescribed fire are often used to provide open areas and suppress woody vegetation (refer to Supporting Information, 'Additional detail about controlling woody vegetation…' section).

IMPLEMENTING KING RAIL CONSERVATION EFFORTS THROUGHOUT THE MAV

The management practices at Choctaw West, which have successfully provided King Rail habitat, could likely be used to inform wetland management and restoration efforts elsewhere in the region. However, the MAV is a large and varied area where a range of factors are likely important when evaluating a site's potential for King Rail use and when deciding which management practices would be most effective. One factor to consider is hydrology (Brewer et al., 2023; Pickens & King, 2013), which includes: the ability to manage water levels and maintain water in wetlands even during droughts, the source of the water, and potential for pesticide runoff into the wetland. Other factors to consider include surrounding land use and management practices, proximity to other emergent wetland areas and rivers, and how far north in the region the site is positioned (Bolenbaugh et al., 2012). Although most of the King Rails at Choctaw West are resident, it is unknown whether King Rails further north in the MAV commonly display this non-migratory behaviour. Research at Choctaw West (Schmit, 2024) indicates that for sites in the southern part of the MAV (southern Arkansas and further south), managers wishing to create conditions suitable for King Rails could consider year-round habitat needs, rather than focusing solely on the breeding or migratory periods.

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In addition to identifying locations well suited to sustain King Rails and effective practices for creating habitat, a general understanding of the factors contributing to King Rail declines in the MAV could help inform conservation actions. During the process of eliciting information to create the PRBN models used in this study, 11 King Rail experts who had managed habitat, or studied King Rails, for ≥3 years in or near the MAV were interviewed. Efforts were made to include a variety of perspectives, from individuals associated with different agencies and universities (Brewer et al., 2025). During the expert interviews (one-on-one conversations), each expert answered the following question: 'What specifically do you think are the primary causes for King Rail population decline in the MAV?' The most common elements of answers included suggestions that current methods of implementing waterfowl habitat management (e.g. a focus on promoting annual plants) are not aligned well with King Rail life-history/habitat requirements ($n = 5$) and that 'loss of habitat,' specifically perennial emergent vegetation ($n = 5$), was problematic. Changes in ditch management (less emergent vegetation retained; $n = 3$) and hydrological change (water too deep or too shallow; $n = 3$) were also common responses to this question. Nearly all the experts ($n = 10$) indicated during the broader conversation that waterfowl management can negatively impact King Rails due to hydrologic modification (e.g. wetland draining during breeding period) and suppression of robust and perennial herbaceous emergent vegetation, like cattails (*Typha* spp.). Most ($n = 8$) experts indicated that loss of habitat/dense emergent vegetation in the MAV has contributed to King Rail population declines. One expert suggested that King Rail conservation efforts could be compatible with waterfowl management, which is in accord with the notion that some aspects of waterfowl management may benefit secretive marsh birds (Malone et al., 2023).

Efforts to increase King Rail populations in the MAV, and beyond, will likely require coordination between researchers, managers and policymakers. The properties predicted herein as being most suitable for King Rails could be prioritized for a coordinated effort to increase understanding of King Rail occupancy and to, if necessary, expand efforts/capability to create and/or maintain habitat via management approaches that we described. Such efforts could also extend to private lands, particularly those in the WRE programme. Our predicted King Rail occupancy results were especially influenced by the presence of other secretive marsh birds (specifically Virginia Rail; *Rallus limicola*)—underscoring the importance of monitoring—and coverage by herbaceous emergent vegetation (Tables S1 and S2; Brewer et al., 2025). Model accuracy would likely benefit from creation of spatial data layers that identify dense herbaceous vegetation which emerges from shallow (<20 cm) water during much of the year. Our knowledge of King Rail habitat in the MAV remains incomplete and predictions from our models have yet to be rigorously field tested. However, timely efforts to conserve King Rails in the MAV based upon available information could help to avoid further population decline—and potentially federal listing—of this species while diversifying habitat conditions within wetland ecosystems across the MAV.

AUTHOR CONTRIBUTIONS

Dustin E. Brewer conceived the idea, collected data, completed analyses and wrote the initial manuscript. Elisabeth B. Webb, Anne E. Mini and S. Keith McKnight helped to conceive the idea, acquire funding and provided edits. Jessica Novobilsky and Jason McCallie collected data and provided edits. Caleb Roberts and Auriel M.V. Fournier helped to acquire funding and provided edits.

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CONFLICT OF INTEREST STATEMENT

We have no conflicts of interest.

PEER REVIEW

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1002/2688-8319.70233.

DATA AVAILABILITY STATEMENT

Maps of predicted King Rail occupancy probabilities (.asc files) are available via Zenodo: https://doi.org/10.5281/zenodo.18912957 (Brewer et al., 2026).

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REFERENCES

Bolenbaugh, J. R., Cooper, T., Brady, R. S., Willard, K. L., & Krementz, D. G. (2012). Population status and habitat associations of the King Rail in the midwestern United States. *Waterbirds*, *35*(4), 535–545. https://doi.org/10.1675/063.035.0404

Brewer, D. E., Gehring, T. M., Garcia, M. M., Shirkey, B. T., Simpson, J. W., & Fournier, A. M. V. (2023). King rail (*Rallus elegans*) home range and microhabitat characteristics in western Lake Erie coastal marshes. *Ecology and Evolution*, *13*(4), e10043. https://doi.org/10.1002/ece3.10043

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Brewer, D. E., Webb, E. B., McCallie, J., Novobilsky, J., Fournier, A. M. V., Roberts, C. P., Mini, A. E., & McKnight, S. K. (2026). Locations and management strategies for prioritizing King Rail conservation efforts in the Mississippi Alluvial Valley [Dataset]. In Ecological Solutions and Evidence. *Zenodo*. https://doi.org/10.5281/zenodo.18912957

Brewer, D. E., Webb, E. B., Mini, A. E., & McKnight, K. (2025). Predicting secretive species distribution using Bayesian networks with and without expert collaboration: A case study incorporating double-blind peer review. *Ecological Solutions and Evidence*, *6*(4), e70140. https://doi.org/10.1002/2688-8319.70140

Budd, M. J., & Krementz, D. G. (2011). Status and distribution of breeding secretive marshbirds in the Delta of Arkansas. *Southeastern Naturalist*, *10*(4), 687–702.

Cooper, T. (2008). King rail (*Rallus elegans*) conservation plan version 1. US Fish and Wildlife Service.

Darrah, A. J., & Krementz, D. G. (2011). Habitat use of nesting and brood-rearing King Rails in the Illinois and Upper Mississippi River valleys. *Waterbirds*, *34*(2), 160–167.

Hargrove, G. M., & Osborne, D. C. (2023). First documentation of King Rail (*Rallus elegans*) reproduction in Arkansas since 2006. *Waterbirds*, *45*(3), 350–353.

Hostetler, J. A., Sauer, J. R., Hines, J. E., Ziolkowski, D., & Lutmerding, M. (2023). The North American breeding bird survey, analysis results 1966-2022: U.S. Geological Survey data release. https://doi.org/10.5066/P9IU1UZ6

LMVJV. (2023). *Secretive marsh bird breeding population and habitat objectives summary*. Lower Mississppi Valley Joint Venture.

Malone, K. M., Webb, E. B., Mengel, D. C., Kearns, L. J., McKellar, A. E., Matteson, S. W., & Williams, B. R. (2023). Wetland management practices and secretive marsh bird habitat in the Mississippi Flyway: A review. *The Journal of Wildlife Management*, *87*(7), e22451. https://doi.org/10.1002/jwmg.22451

Meanley, B. (1969). *Natural history of the King Rail. North American Fauna (#67)*. Bureau of Sport Fisheries and Wildlife.

Pickens, B. A., & King, S. L. (2013). Microhabitat selection, demography and correlates of home range size for the King Rail (*Rallus elegans*). *Waterbirds*, *36*(3), 319–329. https://doi.org/10.1675/063.036.0309

Roberts, C. P., Novobilsky, J., Akpejeluh, P., Berry, L. L., Budd, M. J., & Ferrara, M. C. (2025). *Rallus elegans* (king Rail) occupancy is stable, but habitat is in short supply in the Arkansas Delta. *Ornithological Applications*, *128*, duaf061. https://doi.org/10.1093/ornithapp/duaf061

Rosenberg, K. V., Dokter, A. M., Blancher, P. J., Sauer, J. R., Smith, A. C., Smith, P. A., Stanton, J. C., Panjabi, A., Helft, L., Parr, M., & Marra, P. P. (2019). Decline of the North American avifauna. *Science*, *366*(6461), 120–124. https://doi.org/10.1126/science.aaw1313

Schmit, J. (2024). Migration and breeding ecology of King Rails (*Rallus elegans*) in southeastern Arkansas. Master's Thesis. University of Arkansas.

Smith, C. S., Howes, A. L., Price, B., & McAlpine, C. A. (2007). Using a Bayesian belief network to predict suitable habitat of an endangered mammal—The Julia Creek dunnart (*Sminthopsis douglasi*). *Biological Conservation*, *139*(3–4), 333–347. https://doi.org/10.1016/j.biocon.2007.06.025

Tiner, R. (1984). *Wetlands of the United States: Current status and recent trends*. National Wetlands Inventory, Fish and Wildlife Service, US Department of the Interior.

USGS GAP. (2024). Protected areas database of the United States (PAD-US) 4.0: U.S. Geological Survey data release [Dataset].

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. An example of the mapped output from our Bayesian network (BN) models. These maps represent a high probability of King Rail occupancy (brighter colours) at a wetland complex in Missouri during the breeding period based on a BN model informed by a literature review (LRBN; top panel) and by a BN model informed by a double-blind peer review process involving experts (PRBN; bottom panel). The .asc files that we have provided (see Zenodo) allow for examination of model output (from both models) in any part of the Mississippi Alluvial Valley during the breeding and non-breeding period.

Figure S2. A tractor pulling a cleated rice stubble roller to create patch heterogeneity and plant diversity in a marsh unit. Photo credit: NRCS.

Table S1. Sensitivity of predicted King Rail occupancy probability to parentless nodes for the Bayesian network informed by a literature review during the breeding period. Refer to Brewer et al. (2025) for an overview of what each node represents. Greater values for Mutual Info (entropy reduction) indicate a greater influence on predicted King Rail occupancy probability.

Table S2. Sensitivity of predicted King Rail occupancy probability to parentless nodes for the Bayesian network informed by a literature review during the non-breeding period. Refer to Brewer et al., 2025 for an overview of what each node represents. Greater values for Mutual Info (entropy reduction) indicate a greater influence on predicted King Rail occupancy probability.

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